Completeness- and Update-Tailored Database Design

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***Abstract*—Normalization judges a relational design by the data redundancy it admits, not by how often the application updates the redundant values. Yet every update of a value that a business rule determines rewrites all its redundant occurrences, and a few rules receive most of the updates. Hence two designs that no classical normal form separates can differ without bound in what one workload costs them. Redundancy has a second cost on the incomplete relations of operational databases, whose records are completed as the business process advances. Filling in a missing value can surface a conflict that earlier updates had hidden behind it. When the new value prevails, repairing the conflict rewrites the same redundant occurrences, so a frequently updated rule stored redundantly is paid for twice. We therefore make update frequencies a declared requirement of logical design, next to the completeness requirements that decide on which tuples each rule applies. The heat of a schema is the total update frequency of the non-key functional dependencies it stores. Bounding heat yields a family of normal forms from E-BCNF, the completeness-tailored BCNF, at level zero to E-3NF at the top. Heat is also monotone in the attribute set, which lets our synthesis algorithm return a lossless, dependency-preserving E-3NF decomposition whose hottest subschema is as cool as in any such decomposition. Experiments on a delivery schema show that the update frequencies, and not the structure, decide which design is cheaper, and that choosing by structure can cost more than an order of magnitude. On twelve real-world incomplete relations, our synthesis cools the hottest subschema of about half with at most a few percent more subschemata, and on a release window of another, its design runs the updates that the relation's rules govern fastest.**

***Index Terms*—database design, functional dependencies, incomplete data, normalization, update workloads**

I. Introduction

Business rules enter a relational schema as functional dependencies (FDs), and a decomposition decides how each of them is stored. A rule $X \to A$ stored in a subschema of which X is a key keeps one occurrence per X-value, and an update of its A-value writes one row. Stored as a non-key FD, the rule is enforced by a trigger rather than a key [1], its A-value is repeated in every tuple with the same X-value, and an update rewrites this whole group. The cost of maintaining a rule is therefore the product of two quantities: how often the application updates the values the rule governs, and how many redundant occurrences each update must touch. The first is set by the application, and, like the access frequencies of records [2]–[4], it is skewed: a few rules receive most of the updates while the majority stays cold. Normal forms address only the second. Hence a design chosen on structure alone can store the hottest rule as a non-key FD, where every routine update rewrites a group instead of one row. This paper makes update frequencies a declared requirement of logical design.

Practitioners state these frequencies before the schema exists. The specification of TPC-C, for example, requires at least 43% of its transactions to be payments, each of which updates the balances of a warehouse, a district, and a customer, while no transaction updates a name or an address [5, Cls. 2 and 5.2.3]. Such frequencies have driven physical design for four decades [6], [7], and the design advisors of commercial systems take a workload of statements and their frequencies as input [8], [9]. Physical design, however, leaves the redundancy of the schema in place, so it can speed up an update but not reduce the rows it rewrites. In logical design, frequencies have entered cost models of how far to normalize [10], [11], but not the normal forms and synthesis algorithms that settle the schema, whose criteria are structural. Even *structure-optimal* synthesis, which breaks ties between 3NF schemata by the number of non-key FDs in a cover and then by the number of minimal keys [12], counts the non-key FDs without asking how often each one is updated. Cardinality-bounded design [13], [14] bounds the other quantity, how many occurrences an update touches.

Why incompleteness matters. Operational databases hold incomplete relations [15], whose records are created incomplete and completed as the business process advances. TPC-C, for example, enters every new order without a carrier and every order line without a delivery date, and its Delivery transaction, at least 4% of all transactions, fills both in [5, Cls. 2.4.2, 2.7.4, and 5.2.3]. TPC-C even constrains which values are missing: an order line lacks a delivery date if and only if its order lacks a carrier [5, Cl. 3.3.2.7]. Missing values are thus part of the specification, and filling them in is part of the workload. No design theory over complete relations can express such updates. Over incomplete relations, completeness-tailored design embeds the completeness requirements of an application in the declaration of its constraints: an embedded FD or uniqueness constraint (eFD, eUC) names a set of attributes, its *embedding*, on which tuples must be complete for the rule to apply [16], [17]. E-BCNF and E-3NF grade a design by the data redundancy it admits on the tuples that are complete on an embedding E [17], but not by how often that redundancy is updated.

Like completeness, update frequencies are a requirement of

Full version. The full version of this paper, with an appendix that gives all proofs, is in the repository https://github.com/zhuoxingzhang/completeness-update-tailored-design.

TABLE I: A delivery snapshot r: the scope r^E above the line, the redundant occurrences of the hot rule $z \to c$ in bold, and the pending tuple t_6 below it, outside the scope.

	b(ranch)	c(ourier)	d(ay)	g(ps)	r(ound)	v(an)	z(one)
t_1	North	**Maggie**	Mon	Telco-A	R1	V7	Z1
t_2	South	**Maggie**	Mon	Telco-B	R2	V9	Z1
t_3	North	**Maggie**	Wed	Telco-A	R1	V7	Z1
t_4	North	Maggie	Tue	Telco-A	R1	V7	Z2
t_5	North	Omar	Mon	Telco-B	R2	V4	Z3
t_6	East	Ravi	Tue	$\perp$	R3	V8	Z1

the application, and missing values change what updates cost in three ways. Firstly, the frequency and the redundancy of a rule are both confined to its *scope*, the tuples complete on its embedding, and a *pending* tuple outside the scope needs no maintenance until it enters. Secondly, besides changing a stored value, which we call a *refresh*, an update may fill in a missing value, which we call a *completion*. A tuple that a completion brings into the scope of a rule may disagree with the tuples already there, in a conflict that its null marker $\perp$ had hidden. A refresh changes values only within the scope, so a hot rule leaves such conflicts behind at the rate at which it is refreshed. When the completed value prevails, the redundancy that decides the cost of a refresh also decides the cost of resolving that conflict, so a hot rule stored as a non-key FD is paid for twice. Thirdly, the scope grows as the data completes [17], and with it the groups of every rule, so a hot rule stored as a non-key FD grows more expensive over time. A design for incomplete data must therefore meet both requirements at once, which neither completeness-tailored design nor frequency-driven physical design does.

Cooling hot rules by decomposition. Of the two quantities, the redundancy is the one the design controls [1], [18]. A lossless, dependency-preserving decomposition into BCNF is not always achievable [19], so some rules remain non-key FDs, and the data decide how often the scope repeats their values. Which rules those are is a choice of the decomposition, and we make it by frequency: the rules left non-key must be the ones the application rarely updates. This is illustrated as follows.

Example 1 (Delivery zones)**.** A courier service records deliveries over schema $R = \{branch, courier, day, gps, round, van, zone\}$, abbreviated $\{b, c, d, g, r, v, z\}$. The GPS network g a van reports through is registered only after its unit first reports in, so a tuple can stay pending with $g = \perp$. The application declares each constraint with the completeness requirement under which it applies, and consolidates its requirements in the design scope $E = R$ of fully complete tuples. The declared set Σ consists of the eFD $(cgz;\ z \to c)$ and, each with embedding equal to its attributes, the rules $bcd \to z$, $bc \to v$, $bv \to c$, $cv \to r$, $rz \to b$, and $v \to g$. A zone has one courier on duty among the deliveries whose van has reported in, and a van has one courier within its branch. Replacing a courier refreshes both rules. Couriers are reassigned whenever shifts change or a driver drops out, while

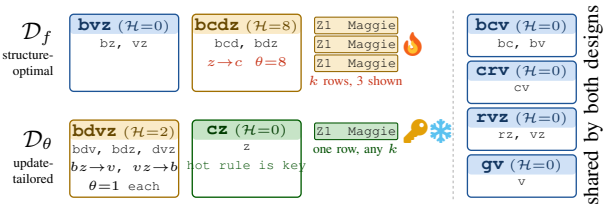


Fig. 1: Two E-3NF decompositions of the delivery schema of Ex. 1, agreeing on four subschemata and parting on two: $\mathcal{D}_f$, returned by structure-optimal synthesis, and $\mathcal{D}_\theta$, returned by ours. Each card is one subschema with its minimal keys and heat $\mathcal{H}$; orange cards maintain non-key FDs.

the van, round, or branch of a delivery is corrected only on occasion. The application therefore declares the update frequency, or *heat*, of the two courier rules as 8 and of every other rule as 1. Table I shows a snapshot. Fig. 1 shows the decompositions $\mathcal{D}_f$ and $\mathcal{D}_\theta$ that structure-optimal synthesis and ours return on one ordering of the input FDs (Ex. 24), both lossless and dependency-preserving over E. Each has five subschemata in E-BCNF and one critical subschema, in E-3NF but not in E-BCNF, which stores one non-key FD in $\mathcal{D}_f$ and two in $\mathcal{D}_\theta$, so structure even prefers $\mathcal{D}_f$. Under the workload they differ sharply. Indeed, $\mathcal{D}_f$ stores the hot rule as a non-key FD on $bcdz$, which repeats the courier of a zone once per branch and day. A reassignment thus rewrites the k rows of the zone in $bcdz$, whereas $\mathcal{D}_\theta$ makes the zone a key of the two-attribute subschema cz and rewrites one row there. A van swap is the mirror image. It rewrites one row per zone of the courier in bvz on $\mathcal{D}_f$ and one row per day in $bdvz$ on $\mathcal{D}_\theta$, so the update frequencies, and not the structure, decide which design is cheaper.

The same redundancy prices the repair of a conflict.

Example 2 (Completions surface latent conflicts)**.** The pending tuple t_6 of Table I claims courier *Ravi* for zone *Z1*, whose scope tuples t_1–t_3 all carry *Maggie*. The disagreement is invisible while g is missing. When the unit of the van first reports in, its network is registered and t_6 enters the scope. The report shows who actually drove, so the newcomer prevails: the zone must be reassigned to *Ravi*, and with it zone *Z2*, which its courier serves with the same van and branch. On $\mathcal{D}_f$ the repair rewrites the k rows of *Z1* in $bcdz$ and those of *Z2*, four in the snapshot, and on $\mathcal{D}_\theta$ one row per zone in the key subschema cz, two at any k. On $\mathcal{D}_f$ the hot rule is thus paid for twice, by every reassignment and every repair.

Contributions. We develop the first design framework that tailors schemata simultaneously to the completeness requirements and the update frequencies of an application. Over complete relations every scope is the whole relation, and the framework tailors 3NF synthesis to update frequencies alone.

1) *Conceptually*, we identify the update workload as a second declared requirement of logical design, next to completeness, and show that frequencies declared under different

embeddings add up per rule on the design scope. We also classify the updates of incomplete data and show that retractions, which replace a value by $\perp$, need no maintenance, while completions alone surface the conflicts refreshes leave hidden outside the scope.

2) *Computationally*, we price a schema by its heat, the total update frequency of the non-key FDs it stores, and bound heat by a family of normal forms from E-BCNF, the anomaly-free level zero, to E-3NF. Heat is monotone in the attribute set and separates designs of equal size by an unbounded factor. Recognizing a level is NP-hard, as recognizing E-3NF is, so the frequencies add no hardness of their own.
3) *Algorithmically*, we establish a synthesis algorithm that returns a lossless, dependency-preserving E-3NF decomposition over the scope whose maximal subschema heat is minimal among all such decompositions, in time polynomial in the number of implied FDs with minimal left-hand sides.
4) *Empirically*, we show on a delivery schema that the update frequencies, and not the structure, decide which design is cheaper over a maintenance window, and that synthesis reaches the cheaper design in all three windows only when the declared heat follows the measured rates. On the schemata of twelve real-world incomplete relations, it cools the hottest subschema of about half with at most a few percent more subschemata, and on a real release window, with heat mined from the release, its design runs the updates that the mined FDs govern fastest.

Organization. Sec. II recalls completeness-tailored design, Sec. III develops the update taxonomy, heat, and the normal forms, and Secs. IV and V present synthesis and completions. Secs. VI–VIII report experiments, related work, and conclusions, and the appendix of the full version gives all proofs.

II. Preliminaries

We recall incomplete relations, embedded constraints, and the completeness-tailored normal forms of Wei and Link [17]. A *relation* r over a finite set R of attributes is a finite set of tuples t that assign each $a \in R$ a value $t(a) \in \mathrm{dom}(a) \cup \{\perp\}$, where the null marker $\perp$ is read as *no information* [20], [21]. A tuple t is *X-total* if $t(a) \neq \perp$ for all $a \in X$.

An *embedded functional dependency* (eFD) over R is an expression $(E; X \to Y)$ with $XY \subseteq E \subseteq R$, and an *embedded uniqueness constraint* (eUC) is an expression $(E; X)$ with $X \subseteq E \subseteq R$ [16], [17]. The *scope* of an embedding E in r is $r^E = \{t \in r \mid t \text{ is } E\text{-total}\}$. A relation r satisfies the eFD $(E; X \to Y)$ if every pair of tuples in r^E that agrees on X agrees on Y, and it satisfies the eUC $(E; X)$ if no two distinct tuples in r^E agree on X. Over total relations every scope is the whole relation, and eFDs and eUCs are FDs and UCs.

An application declares a set Σ of eFDs and eUCs and consolidates its completeness requirements in a single *design scope* $E \subseteq R$ [17]. The *reduct* of Σ to E is the set $\Sigma[E]$ of ordinary FDs $X \to Y$ and UCs X arising from members of Σ whose embedding is contained in E. On the scope, where every tuple is E-total, these members hold exactly when $\Sigma[E]$ holds classically, so design over the scope is classical design for $\Sigma[E]$ [17]. A schema (R, Σ) with design scope E is in *E-BCNF* (*E-3NF*) when $(R, \Sigma[E])$ is in BCNF (3NF), and E-losslessness and E-dependency-preservation are losslessness and dependency-preservation with respect to r^E [17]. For $S \subseteq R$, the constraints Σ_S of the subschema S are the FDs over S that $\Sigma[E]$ implies, and the subschema is in E-BCNF (E-3NF) when (S, Σ_S) is in BCNF (3NF).

A design stores r^E projected onto its subschemata and the remaining tuples in a full-width *remainder* [17]. The subschemata enforce $\Sigma[E]$ on the scope, and an eFD whose embedding is a proper subset of E is checked on a tuple of the remainder once that tuple becomes E-total. E-BCNF and E-3NF grade such a design by the redundancy it admits on the scope. Sec. III prices that redundancy by how often the application updates it.

III. Updates, Heat, and Tailored Normal Forms

Throughout, fix a schema (R, Σ) with design scope E. This section classifies the updates of incomplete relations, attaches declared frequencies to the constraints they refresh, defines the heat of a schema, bounds it by a family of normal forms justified by update anomalies, and settles their complexity.

A. A Taxonomy of Updates over Incomplete Data

Updates over incomplete relations differ from updates over complete ones in one respect: a value can move into or out of $\perp$, and thereby a tuple can enter or leave a scope.

Definition 3 (Update taxonomy)**.** Let r be a relation over R, $t \in r$, $a \in R$, and let t' result from t by replacing $t(a)$ with v. The replacement is a *refresh* if $t(a) \neq \perp$ and $v \neq \perp$, a *completion* if $t(a) = \perp$ and $v \neq \perp$, and a *retraction* if $t(a) \neq \perp$ and $v = \perp$.

Retractions need no constraint maintenance, since they only shrink scopes: a retracted tuple either leaves a scope or keeps the projection it had on it, so every pair of tuples that a constraint still sees is a pair it saw before. Completions, conversely, grow scopes and can therefore cause violations, whose cost Sec. V analyzes. Inserts reduce to these two cases. An insert of an E-total tuple enters the scope and costs what a completion costs. An insert of a tuple with a missing value in E lands outside the scope and, like a retraction or a deletion, needs no maintenance until a completion moves it in. The rest of this section studies refreshes, the routine updates.

B. Update Modes and Heat

Applications refresh the values their business rules determine, so we declare update behavior on the constraints, as completeness requirements are declared on them [17].

Definition 4 (Update mode)**.** An *update mode* over R is an expression $\mu = (E'; X \to A)$ with $XA \subseteq E' \subseteq R$. An application of μ to r picks an X-total condition tuple x and a

value $v \in \mathrm{dom}(A)$, and refreshes $t(A) := v$ for every $t \in r^{E'}$ with $t[X] = x[X]$.

The tuples refreshed form the X-*group* of x in $r^{E'}$, the tuples with matching values on X. Every declared eFD $(E'; X \to Y) \in \Sigma$ induces one mode per attribute of Y, and M_Σ denotes the set of these modes. How often each of them is applied is the second requirement the application declares.

Definition 5 (Heat map)**.** A *heat map* is a function θ assigning each declared mode of M_Σ a positive heat, the update frequency the application declares for it on a common scale.

Like the completeness requirement E, the heat map is stated by the application and not read off an instance. All results hold for arbitrary positive heats, and only their ratios matter, since schemata are compared by sums of heats (Def. 9 below). A mode without a stated frequency has the default heat 1, the least a declaration can state, so no rule is declared away. A declared mode names one rule, but several rules of the reduct can determine one attribute, as $z \to c$ and $bv \to c$ determine the courier, and a refresh has to keep all of them satisfied.

Lemma 6 (Co-refreshed rules)**.** *Let r^E satisfy $\Sigma[E]$, let v be* fresh *in that it occurs in no a-value of r^E, and let r' arise from r^E by rewriting the a-value of the tuples of a set T to v. Then r' satisfies every non-trivial FD $X \to a$ that $\Sigma[E]$ implies if and only if T is a union of* components *of a, the classes of the finest partition of r^E in which every X-group of every such FD lies within one class.*

A refresh therefore rewrites a component, whatever rule it is issued against, and moves a whole group of every rule that determines the attribute.

Example 7 (The component of a reassignment)**.** In Table I, zone *Z1* is reassigned to a new courier Ada. The mode $z \to c$ names t_1–t_3, but t_1 and t_4 share branch $North$ and van $V7$, so $bv \to c$ carries t_4 along and the component is t_1–t_4: zone *Z2* moves with *Z1*, since its courier drives the same van from the same branch. Rewriting t_1–t_3 alone violates $bv \to c$, so both courier rules are refreshed.

Conversely, several declared modes can share one FD part $X \to A$, and on the reduct they refresh the same tuples.

Lemma 8 (Heat aggregation)**.** *Let $(E_i; X \to A)$, for $i = 1, \ldots, m$, be the modes in M_Σ with embedding contained in E and identical FD part $X \to A$. On the scope, an application of any of these modes with condition x refreshes exactly the tuples of r^E with X-value $x[X]$, whatever the mode's embedding. Hence the frequency with which $X \to A$ is refreshed on the scope is the sum of the modes' frequencies.*

A design is therefore priced by one frequency per rule of the reduct, whatever embeddings the application declared. From now on every declared mode is embedded within E, since a mode with $E' \not\subseteq E$ concerns designs over several scopes (Sec. VIII). Refreshes of pending tuples in the remainder cost the same on every design and are left out.

Applying a mode to a design. A design is maintained subschema by subschema, each under its own constraints, as in the update studies of structure-optimal design [1], [12]. On a decomposition, an application of a mode $X \to A$ rewrites the A-values of the component that contains its X-group (Lem. 6), and every subschema is then kept equal to the projection of the refreshed scope. A subschema that contains XA rewrites the rows of the group, one row where X is a key of it. A subschema that contains A but not all of X cannot locate the group and replaces the projections of the component's tuples, one row per distinct projection. Heat prices only the first kind, where the design decides between one row and the group.

C. The Heat of a Schema

A design stores an FD redundantly whenever dependency preservation over the scope leaves no cover in which the FD is a key dependency [17], [22]. Which FDs a schema stores as non-key is decided by its attribute set: structure-optimal design counts how few of them need a trigger of their own [12], and we count how often all of them are refreshed.

The rules a design can store are the members of the *atomic closure* $\mathcal{A}$ of $\Sigma[E]$, the non-trivial FDs $X \to a$ that $\Sigma[E]$ implies and whose determinant X is minimal [23], [24], so heat is assigned to them. The *aggregated map* θ_E assigns each declared member of $\mathcal{A}$ the sum of the heats of its modes (Lem. 8), and a member that no declared mode refreshes the default heat 1. A declared mode $X \to a$ whose determinant is not minimal charges every member $Y \to a$ with $Y \subsetneq X$, whose groups contain those of X.

Definition 9 (Heat of a schema)**.** The *heat* of a schema (R_i, Σ_i) with atomic closure $\mathcal{A}_i$ is $\mathcal{H}_\theta(R_i) = \sum \{\theta_E(X \to a) \mid X \to a \in \mathcal{A}_i,\ X \text{ not a superkey of } R_i\}$.

Keys cost no heat, since in a subschema of which X is a key the group is one row and its refresh is a plain write. A refresh of a non-key FD rewrites every row of its group, and it fires a trigger even if the FD is not in the cover that the triggers enforce, namely the trigger of the cover FD through which it is derived. Heat therefore counts every non-key FD, whether in that cover or not.

Example 10 (Heat of the delivery subschemata)**.** The atomic closure of the delivery reduct of Ex. 1 adds eleven implied FDs to the declared rules, among them $bz \to v$, $vz \to b$, and $bdv \to z$. The minimal keys of R are bcd, bdv, bdz, drz, and dvz. The undecomposed schema R stores 16 non-key FDs, the two that determine the courier at the declared heat 8 and the others at their default heat 1, so $\mathcal{H}_\theta(R) = 30$. Subschema cz of $\mathcal{D}_\theta$ has the key z and no non-key FD, so its heat is 0. Subschema $bcdz$ of $\mathcal{D}_f$ has the keys bcd and bdz and the non-key FD $z \to c$, so $\mathcal{H}_\theta(bcdz) = 8$. Subschema $bdvz$ of $\mathcal{D}_\theta$ has the non-key FDs $bz \to v$ and $vz \to b$, so $\mathcal{H}_\theta(bdvz) = 2$.

That both critical subschemata stay below the heat of R is no coincidence: adding attributes can only add non-key FDs.

Proposition 11 (Monotonicity)**.** *If $S \subseteq S'$ are schemata over R, then $\mathcal{H}_\theta(S) \leq \mathcal{H}_\theta(S')$.*

By monotonicity, synthesis attains the least heat of any E-lossless, E-dependency-preserving E-3NF design (Sec. IV).

D. A Family of Tailored Normal Forms

Bounding heat yields a family of normal forms.

Definition 12 ((h, E)-tailored normal form)**.** Let $h \geq 0$ be a *level* of heat. A schema (R_i, Σ_i) is in (h, E)-*tailored normal form*, or (h, E)-TNF, if it is in E-3NF and $\mathcal{H}_\theta(R_i) \leq h$. A decomposition is in (h, E)-TNF if all its subschemata are. Its *heat* is the maximum subschema heat.

Heat weighs a rule by its declared frequency alone. The group sizes that multiply each frequency belong to the instance, which does not exist when the schema is designed and grows as the scope fills (Sec. V), and a design is chosen once, for every instance, so it cannot be weighted by them. Where the application can bound the groups of a rule, it declares a cardinality constraint [14] (Sec. VIII). A decomposition is weighed by its hottest subschema, not by the sum over its subschemata. By Prop. 11 no subschema is hotter than R, so decomposing never raises the level, and synthesis, which examines the hottest critical schemata first, drives it to the least value any cover allows (Sec. IV).

Its extremes are the classical normal forms.

Proposition 13. *A schema is in E-BCNF if and only if its heat is* 0*, that is, if and only if it is in* $(0, E)$-TNF*.*

The levels are nested and topped by E-3NF.

Proposition 14. *For* $h \leq h'$*, every schema in* (h, E)-TNF *is in* (h', E)-TNF*, and* $(\mathcal{H}_{\text{tot}}, E)$-TNF*, with the per-schema bound* $\mathcal{H}_{\text{tot}} = \sum_{\sigma \in \mathcal{A}_i} \theta_E(\sigma)$*, coincides with E-3NF. Every step of the hierarchy is strict for suitable* Σ *and* θ*.*

Indeed, $\{a, b, c\}$ with the rules $ab \to c$ and $c \to b$ is in (h, E)-TNF exactly when h is at least the heat of $c \to b$.

Under uniform heat, the family degenerates to structure.

Proposition 15 (Uniform heat degenerates to structure)**.** *If* θ_E *is constant, the heat of a schema is proportional to its number of non-key atomic FDs, and minimizing decomposition heat minimizes the maximal number of such FDs per subschema.*

Note that heat counts all non-key atomic FDs, while structure-optimal synthesis counts the non-key FDs of one cover [12]. The two counts agree wherever no non-key atomic FD is implied by the other atomic FDs, as on the two subschemata on which the designs of Ex. 1 differ.

Under skew, however, the family separates designs of equal size, and no structural measure bounds that separation.

Proposition 16 (Unbounded separation)**.** *For every* L *there are* (R, Σ, E, θ) *with two E-3NF decompositions that are lossless and dependency-preserving over* E*, of equal size, and whose heats differ by a factor of* L*. The design of larger heat stores one non-key FD fewer than the other.*

The running example illustrates both propositions.

Example 17 (Levels of the delivery designs)**.** By the heats of Ex. 10, the designs of Fig. 1 sit at levels 8 and 2: $\mathcal{D}_f$ is in $(8, E)$-TNF and in no lower level, and $\mathcal{D}_\theta$ is in $(2, E)$-TNF. Both are in E-3NF and neither is in E-BCNF, so the classical normal forms do not separate them, and the count of non-key FDs, which a uniform declaration measures (Prop. 15), prefers $\mathcal{D}_f$. Declaring $z \to c$ at heat $2L$ instead of 8 raises the level of $\mathcal{D}_f$ to $2L$ and leaves that of $\mathcal{D}_\theta$ at 2, which is the separation of Prop. 16 on this schema.

E. Justification by Update Anomalies

Normal forms are justified by the anomalies they characterize [18], [25], [26], and E-BCNF by E-redundancy [17]. We justify the heat hierarchy by the updates it forces. Subschemata generated from the reduct lie within E and store projections of the scope, so their instances are total relations.

Definition 18 (Anomalous application)**.** An application of a mode $X \to A$, with condition x and a fresh value v, is *anomalous* on a schema instance $r_i \models \Sigma_i$ if rewriting the A-value of a single tuple of the X-group of x violates Σ_i, while rewriting the whole group does not.

An anomalous application cannot be confined to one tuple, since satisfaction forces the write onto the whole group.

Proposition 19 (Anomaly boundary)**.** *A schema admits an anomalous application on some schema instance if and only if it is not in E-BCNF.*

The levels above $h = 0$ measure what enforcing the unavoidable anomalies costs. A schema is enforced by declaring its minimal keys as uniqueness constraints and a cover of its non-key FDs as triggers on writes to their attributes [1]. The stored tuples are total, so each uniqueness constraint is enforced as a *certain key* by a plain unique index [27], [28]. Hence heat counts the refreshes that enforcement leaves to triggers rather than keys, once per non-key FD a refresh moves, and level h bounds it.

The bound extends to completions: one that enters the scope with a fresh conflicting value forces the write pattern of Def. 18 on every subschema that stores the rule as a non-key FD (Prop. 27), so the anomalies a level tolerates are also the repairs that completions can force.

F. Complexity

The framework poses two problems: recognizing the level of a given schema, and deciding whether a design at a given level exists, which at level 0 is the classical question of a lossless, dependency-preserving decomposition into BCNF [19].

Theorem 20 (Determination)**.** *Given a schema with a cover of its projected constraints* Σ_i*, a heat map, and a bound* h*, deciding membership in* (h, E)-TNF *is NP-hard, already under constant heat. If the atomic closure* $\mathcal{A}_i$ *is computable in polynomial time, the problem is in NP and interreducible with deciding E-3NF. Given* (R, Σ)*,* E*, a heat map, and* h*, deciding whether an E-lossless, E-dependency-preserving*

decomposition into (h, E)-TNF exists is NP-hard, already for $h = 0$ under constant heat.

The frequencies add no hardness of their own: recognition is hard already for 3NF [29], and the design problem already at level 0. Synthesis, however, need not recognize levels: given $\mathcal{A}_i$, heat costs one attribute closure per member.

IV. Heat-Aware Synthesis

We now determine the minimum heat over all E-lossless, E-dependency-preserving, E-3NF decompositions and give a synthesis algorithm that attains it.

A decomposition is E-dependency-preserving exactly when the members of $\mathcal{A}$ that lie within its subschemata imply $\mathcal{A}$. Every E-3NF schema that contains the attributes of an atomic FD $X \to a$ contains Xa, which is itself in E-3NF [30], [31], so by Prop. 11 none of them is cooler than Xa.

Lemma 21 (Selection lower bound)**.** *Let h^* be the minimum of $\max_{X \to a \in F} \mathcal{H}_\theta(Xa)$ over all $F \subseteq \mathcal{A}$ that imply $\mathcal{A}$. Every E-lossless, E-dependency-preserving, E-3NF decomposition has maximal subschema heat at least h^*.*

Equivalently, h^* is the least t at which the members of $\mathcal{A}$ whose schemata have heat at most t imply $\mathcal{A}$, so implication tests alone find it. We compute it on the running example.

Example 22 (The delivery optimum)**.** In Ex. 1, every cover retains $z \to c$, $bc \to v$, $bv \to c$, $cv \to r$, and $v \to g$, whose schemata have heat 0, and retains or derives $bcd \to z$. The cheapest derivation retains $bdv \to z$, with $\mathcal{H}_\theta(bdvz) = 2$, while $\mathcal{H}_\theta(bcdz) = 8$. Hence $h^* = 2$, attained by $\mathcal{D}_\theta$.

To reach h^*, Alg. 1 follows classical synthesis [30], [31] and its structure-optimal refinement, which eliminates redundant critical schemata before redundant BCNF ones [12]. It makes two changes: it runs on the reduct, and it eliminates redundant critical schemata in decreasing order of heat rather than of their number of non-key FDs. Redundant E-BCNF schemata, all of heat zero, are eliminated in decreasing number of minimal keys as before. The heat order attains h^* (Thm. 23): the members whose schemata have heat at most h^* imply $\mathcal{A}$ and are examined after every hotter critical schema, which is thus redundant. No structural order can do so (Prop. 16).

Theorem 23 (Correctness and optimality)**.** *Alg. 1 returns a decomposition that is E-lossless, E-dependency-preserving, and in E-3NF, and whose maximal subschema heat is h^*, the minimum over all such decompositions (Lem. 21). Subject to that minimum, the maximal number of minimal keys over its E-BCNF subschemata is minimal among the decompositions obtained by eliminating redundant FDs from the set $\mathcal{A}_w$ that its first pass leaves.*

Thm. 23 uses heat only through Props. 11 and 13, so it holds for every monotone cost that is zero on E-BCNF schemata. We illustrate Alg. 1 by deriving the designs of Fig. 1.

Example 24 (Synthesis on the delivery reduct)**.** We start with the atomic closure of Ex. 10, whose 18 members generate two

Algorithm 1 Tailor(R, Σ, E, θ)

Require: schema R, constraints Σ, design scope E, heat map θ over M_Σ
Ensure: E-lossless, E-dependency-preserving, E-3NF decomposition $\mathcal{D}$ of (R, Σ) over E that minimizes the maximal subschema heat

1: $\mathcal{A} \leftarrow$ atomic closure of $\Sigma[E]$ [23], [24]; $\theta_E \leftarrow$ aggregated map of θ on $\mathcal{A}$; $\mathcal{A}_w \leftarrow \mathcal{A}$; $\mathcal{D} \leftarrow \emptyset$
2: $\mathcal{C} \leftarrow \{X \to a \in \mathcal{A} \mid (Xa, \Sigma_{Xa})$ not in E-BCNF$\}$
3: **for all** $X \to a \in \mathcal{C}$ **do**
4: $\quad \mathcal{M}[X \to a] \leftarrow \mathcal{H}_\theta(Xa)$, the sum of θ_E over the FDs of $\mathcal{A}$ inside Xa that are non-key in Xa
5: **for all** $X \to a \in \mathcal{C}$ in decreasing order of $\mathcal{M}[X \to a]$, ties in the order of $\mathcal{A}$ **do**
6: $\quad$ **if** $\mathcal{A}_w \setminus \{X \to a\} \models X \to a$ **then** $\mathcal{A}_w \leftarrow \mathcal{A}_w \setminus \{X \to a\}$ **else** $\mathcal{D} \leftarrow \mathcal{D} \cup \{(Xa, \Sigma_{Xa})\}$
7: **for all** $X \to a \in \mathcal{A} \setminus \mathcal{C}$ in decreasing order of the number of minimal keys of (Xa, Σ_{Xa}), ties in the order of $\mathcal{A}$ **do**
8: $\quad$ **if** $\mathcal{A}_w \setminus \{X \to a\} \models X \to a$ **then** $\mathcal{A}_w \leftarrow \mathcal{A}_w \setminus \{X \to a\}$ **else** $\mathcal{D} \leftarrow \mathcal{D} \cup \{(Xa, \Sigma_{Xa})\}$
9: $\mathcal{D} \leftarrow \mathcal{D} \setminus \{(S, \Sigma_S) \mid \exists\, (S', \Sigma_{S'}) \in \mathcal{D}$ with $S \subsetneq S'\}$
10: **if** $S \to R \notin \Sigma[E]^+$ for all $(S, \Sigma_S) \in \mathcal{D}$ **then**
11: $\quad \mathcal{D} \leftarrow \mathcal{D} \cup \{(K, \Sigma_K)\}$ for a minimal key K of $(R, \Sigma[E])$ [32]
12: store r^E in $\mathcal{D}$ and $r \setminus r^E$ in a full-width remainder [17]
13: **Return** $\mathcal{D}$

critical schemata, both from a rule that determines the zone. The schema $bcdz$, generated by $bcd \to z$, stores the hot rule $z \to c$ and has heat 8, and $bdvz$, generated by $bdv \to z$, stores $bz \to v$ and $vz \to b$ and has heat 2 (line 4). Line 6 therefore examines $bcd \to z$ first. It is redundant, since $bc \to v$ and $bdv \to z$ imply it, and is removed. The FD $bdv \to z$ is then not redundant, since the remaining FDs derive only c, g, and r from bdv, so $bdvz$ is added to the decomposition. Line 7 examines the E-BCNF schemata by their numbers of minimal keys, 2 for bcv, brz, bvz, and rvz and 1 for the eight others, and on the order we trace retains bcv, rvz, bvz, cz, crv, and gv, the schemata of the FDs that the remaining ones do not imply. Since $bvz \subseteq bdvz$, line 9 removes bvz, and line 11 adds no key, since $bdvz$ contains bdz. The final decomposition is $\mathcal{D}_\theta$, of heat $h^* = 2$. Structure-optimal synthesis examines $bdv \to z$ first, as $bdvz$ stores two non-key FDs and $bcdz$ one. This FD is redundant while $bcd \to z$ is present, and its removal leaves $bcd \to z$ irredundant, so $bcdz$ of heat 8 is added instead. The second pass then runs as before, but line 9 now absorbs cz into $bcdz$ and leaves bvz, which no retained subschema contains, so the result is $\mathcal{D}_f$. Some orders retain brz in place of bvz and rvz, and both algorithms then return that variant, with the same critical subschema. On this pair the count of non-key FDs and the heat point in opposite directions.

Like structure-optimal synthesis [12], Alg. 1 runs in worst-case exponential time, and unavoidably so unless P = NP, since the level it attains decides the design problem of Thm. 20. The only source of exponential cost is the atomic closure on line 1, whose size can be exponential in the size of Σ.

Proposition 25 (Cost of synthesis)**.** *Given the atomic closure $\mathcal{A}$, Alg. 1 runs in time polynomial in $|\mathcal{A}|$ and $|R|$.*

Hence h^* is computable in polynomial time from $\mathcal{A}$.

V. Completions and Deferred Maintenance

Completions move tuples into scopes and make them subject to the constraints. We price the repair of a conflicting completion, trace conflicts to their sources, and name the design property that decides what they cost.

Definition 26 (Completion mode)**.** A *completion mode* over R is an attribute $A \in E$. An application picks a tuple t with $t(A) = \perp$ that is $(E \setminus \{A\})$-total, and replaces $t(A)$ by some $v \in \mathrm{dom}(A)$, moving t into the scope r^E.

On a decomposed design, applying a completion mode is a *scope entry*: the entering tuple's projections are inserted into every subschema, the tuple is removed from the remainder, and all constraints must then hold. The entry check is cheap, while the design decides what repairing a violation costs.

Proposition 27 (Conflict amplification)**.** *Let $X \to A$ hold on the scope, let the entering tuple t disagree on A with the k scope tuples of its X-group, and let the newcomer-wins policy apply. A design that stores $X \to A$ as a non-key FD on a subschema rewrites there one stored A-value per distinct projection of the group, all k of them whenever the subschema separates the group's tuples. A design that stores XA only in subschemata of which X is a key holds the group in one row of each, and rewrites one row in each.*

An earlier refresh can leave such a disagreement behind.

Definition 28 (Latent conflict)**.** Let $X \to A$ hold on the scope r^E of a relation r. An XA-total tuple $t \in r \setminus r^E$ carries a *latent conflict* with $X \to A$ if a tuple of r^E agrees with t on X but not on A.

No constraint sees a latent conflict, whose carrier lies outside the scope, yet it decides the repair a completion needs.

Proposition 29 (Dynamics of latent conflicts)**.** *Let the scope of r satisfy $\Sigma[E]$ and let $X \to A$ hold on it.*

1) *A retraction creates no latent conflict.*
2) *A refresh that moves an X-group of the scope to a fresh A-value turns every XA-total pending tuple that agreed with the group into a carrier of a latent conflict, unless the tuple is total on the embedding of the refreshing mode and is refreshed with the group.*
3) *Only a completion turns a latent conflict into a violation, and Prop. 27 gives the cost of its repair.*
4) *An insert into the scope is a scope entry: a disagreement with its X-group is a violation at once, repaired as in Prop. 27. An insert outside the scope violates no constraint of the scope, and carries a latent conflict with $X \to A$ from the start if and only if it is XA-total and disagrees on A with its X-group.*

Missing values thus postpone maintenance without removing it. The entry check of a completion is blind to heat, since it validates every rule of the reduct once, on the subschema that stores it. Its repair is not: by (2), every refresh of a group with agreeing pending tuples outside the mode's embedding makes their later completions conflicting, so the hotter a rule, the more latent conflicts it creates. The workload decides how many arise, and the design decides what each costs (Prop. 27). Completions also cause *completeness drift*: as values are filled, the groups grow, and both refresh and repair cost more.

Both costs turn on one property of the design. A mode $(E'; X \to A)$ with $E' \subseteq E$ is *key-materialized* in a design if some subschema contains XA and has X as a key. On such a subschema the mode is enforced as a certain key [28]: the entry check of a completion is the lookup that its insert performs anyway, and the repair of Prop. 27 rewrites one row there. Otherwise a trigger enforces the mode on a subschema in which its group occupies many rows, and every conflicting completion pays the amplification there. The running example traces one conflict from its source to its cost.

Example 30 (The life of a latent conflict)**.** The pending tuple t_6 of Table I claims *Ravi* for zone *Z1*, whose scope tuples carry *Maggie*. It acquires that disagreement when the zone is reassigned, since the mode $(cgz;\ z \to c)$ refreshes no tuple whose network is missing (Prop. 29(2)). Only the completion of the network turns the latent conflict into a violation (Prop. 29(3)). The newcomer prevails, as in Ex. 2. The repair then rewrites the k rows that $bcdz$ holds for *Z1* on $\mathcal{D}_f$, and its one row of cz on $\mathcal{D}_\theta$ (Prop. 27). The mode is key-materialized on cz, so $\mathcal{D}_\theta$ finds that row by a key lookup, while $\mathcal{D}_f$ scans the group.

VI. Experiments

A. Research Questions

We structure the evaluation along seven questions.

RQ1 How does the cost of maintaining a redundantly stored rule scale with its redundancy, for refreshes and for completions, across designs?

RQ2 How do classical, structure-optimal, and heat-aware synthesis compare on schema-level metrics, on the redundant value occurrences their outputs store, and on synthesis time?

RQ3 What do workloads that mix refreshes and completions cost as the completion share varies, and does tailoring slow reads?

RQ4 How sensitive are the designs to skew in the heat distribution?

RQ5 How robust are they to misestimated heat and to completeness drift, as scopes and groups grow?

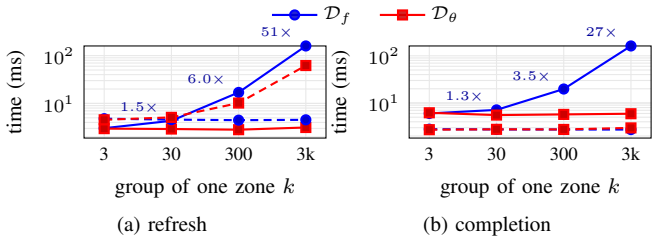
RQ6 What does a whole declared workload cost, once refreshes, completions, retractions, inserts, and deletions arrive together over one maintenance window, each at its own rate?

RQ7 What do the priced updates of a real window cost on the design tailored to that window, once its heat is mined from the release, and as these updates multiply?

The seven questions build one argument: RQ1 establishes the mechanism, RQ2 and RQ3 quantify what synthesis makes

TABLE II: The three workload profiles (RQ4, RQ5, RQ6): the updates of one window of 1,000, by kind.

update	profile		
	delivery desk	split	fleet office
refresh the courier of a zone	300	160	20
refresh the van of a courier	20	160	300
refresh the zone of a delivery	50	50	50
refresh the round of a van	30	30	30
refresh the branch of a round	10	10	10
refresh the network of a van	10	10	10
complete a delivery, clean	120	120	120
complete a delivery, conflicting	20	20	20
retract a value	50	50	50
insert an E-total delivery	200	200	200
insert a pending delivery	100	100	100
delete a delivery	90	90	90



(a) refresh (b) completion

Fig. 2: Scaling the group k of one zone on the delivery designs (RQ1): (a) one reassignment (solid) and one van swap (dashed), (b) one entering tuple, conflicting (solid) and clean (dashed); labels give the time on $\mathcal{D}_f$ over that on $\mathcal{D}_\theta$.

of it on real reducts and under mixed workloads, RQ4 and RQ5 define its limits, and RQ6 runs it live over a whole window. RQ7 then mines a window from a real release, at rates we do not choose, and times the design tailored to that window.

B. Experimental Setup

Environment. The repository named on the first page holds the implementation, the data, the constraint sets, and the protocol behind every number. All algorithms are implemented in Python, and the operational studies run on MySQL, on a workstation with a 12-core Intel Core i7-12700 and 128 GB of memory (Windows 11, MySQL 8.0.29, Python 3.12). Unless stated otherwise, a reported time is the median of three repetitions that visit the designs round-robin. Every design indexes its minimal keys and no other attributes, as in the update studies of structure-optimal design [12].

Datasets. We use 12 real-world incomplete relations standard in data profiling and normalization research [33], [34], of 10 to 38 attributes, 108 to 17,305,799 rows, and 0.2% to 76.2% null cells, with the FD sets mined on them, of 8 to 2,995 FDs. The mining reads a null marker as no information, as the comparisons of SQL do. Each mined set is taken as the reduct $\Sigma[E]$ over the full attribute set.

Baselines. We compare three synthesizers on one front end, which computes the reduct $\Sigma[E]$, its atomic closure, and the keys, so the objective is the only variable. 3NF is the classical lossless, dependency-preserving synthesis [30], [31], SO is structure-optimal synthesis [12] run on the reduct, and HA is heat-aware synthesis by Alg. 1. Redundant value occurrences are counted in the sense of [17], [25].

Heat assignment. The benchmark FD sets come without workload logs, so heat is declared in three ways. Graded heat makes each atomic FD hot with probability p and then draws its level uniformly from $\{1, \ldots, 10\}$. Data-derived heat maps the redundant value occurrences a rule causes proportionally onto the same scale. Under the worst single rule, one rule is hot at level 10 and every other rule cold, and we report the rule on which the two optimizing designs differ most. We sweep p and report means over 20 draws.

Controlled studies. Five studies run on generated instances of the delivery schema of Ex. 1 that satisfy the declared rules and no others. The number of zones is fixed, so growing the instance grows the group k of deliveries a zone carries. Every refresh writes a fresh value (Lem. 6). Table II lists the three workloads under which RQ4, RQ5, and RQ6 issue whole maintenance windows of 1,000 updates, at equal volume so that the mix is what separates them. At the delivery desk most refreshes reassign couriers, at the fleet office most swap vans, and the split profile runs both equally. These studies declare heat by operation, as a log does: an operation of Table II carries its own rate, and every rule that determines the attribute it rewrites carries that rate (Lem. 6). A conflicting completion enters a tuple that disagrees with its group on the attribute the window rewrites most often (Prop. 29(4)), so its rate is added to that of the hottest refresh, and at the split profile, where the two refreshes tie, to the courier's.

Real release. RQ7 uses 23 columns of the COVID table of Our World in Data, which is revised in place in a repository that keeps every edition.[1] The requirement E holds 21 of them, leaving a third of the tuples E-complete. A window is a pair of editions: the relation is the earlier one, the constraints are mined on it, and the workload is the difference between the two, coalesced into the updates that produce it. The heat of a rule $X \to A$ is the number of the coalesced updates that refresh A inside the scope (Lem. 6), and a rule the window does not refresh keeps the default heat 1. An update is priced if some FD of the reduct determines the attribute it refreshes. Every design executes the whole window, and we report the mean time of its priced updates.

Re-numbered levels. A priced attribute with n updates in the scope receives one of ten levels ℓ in proportion to n, and the maps ℓ^2, 2^ℓ, and ℓ^3 re-number these levels to multiply the priced updates. A multiplied update alternates between the released value and the earlier one, so its frequency follows the new level and the window still ends in the released edition.

C. Operational Costs under Controlled Redundancy (RQ1)

RQ1 measures the case study of Exs. 1 and 2 on instances that grow, and Fig. 2 shows that redundancy governs the cost of maintaining a rule stored as a non-key FD. One reassignment

[1] https://ourworldindata.org/coronavirus, https://github.com/owid/covid-19-data

TABLE III: Design heat under the graded declaration at $p = 0.5$ (RQ2), totals rounded, HA bold where strictly coolest; *: SO and HA coincide on every metric at every p.

		maximal			total					maximal			total		
reduct	$\|\mathcal{A}\|$	3NF	SO	HA	3NF	SO	HA	reduct	$\|\mathcal{A}\|$	3NF	SO	HA	3NF	SO	HA
routes*	24	4.2	4.2	4.2	4	4	4	bridges	128	8.2	7.4	**7.3**	30	19	**18**
claims*	28	0.0	0.0	0.0	0	0	0	echo	355	10.5	9.4	**8.6**	36	26	**23**
breast	47	5.8	5.8	**4.4**	8	8	**6**	ncvoter	1,249	20.4	15.4	**12.8**	165	90	**78**
biocase*	47	0.0	0.0	0.0	0	0	0	dblp10k	1,377	18.9	18.9	18.9	286	269	**263**
hospital*	83	7.3	0.0	0.0	9	0	0	hepatitis	4,513	44.6	42.3	**42.2**	2,198	2,009	**1,923**
pdbx*	101	4.8	4.8	4.8	16	16	16	weather	6,582	52.6	36.6	**34.2**	4,039	3,302	**3,110**

rewrites every occurrence a zone carries on $\mathcal{D}_f$ and a constant number of rows on $\mathcal{D}_\theta$, the group rewrite of Sec. III-B. The time follows that count (Fig. 2(a)): the designs tie at the smallest group, where the rewrite is small against the cost of a statement, and part by $51\times$ at the largest. A van swap is the same mechanism with the designs exchanged, so neither design dominates on refreshes. Completions separate into the two kinds of scope entry (Fig. 2(b)). Clean completions write the same rows and take the same time on both designs at every scale, since both issue the same inserts. Conflicting completions repair a latent conflict, which rewrites the same rows as a reassignment and takes up to $27\times$ as long on $\mathcal{D}_f$.

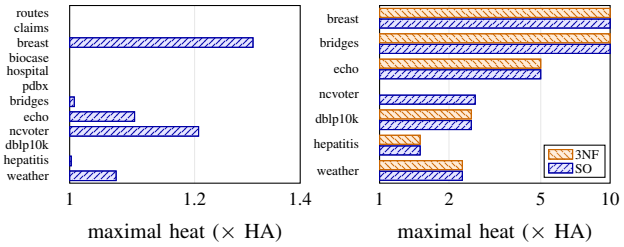
D. Schema-Level Gains on Real Reducts (RQ2)

RQ1 shows that a design sets the cost of a hot rule by whether it stores the rule under a key. RQ2 asks how often synthesis has that choice on the reducts of real relations, and what taking it costs in structure, in stored redundancy, and in time. The three declarations test different things: graded heat asks whether the objective ever loses to structure, data-derived heat lets what a design stores be checked against what it is told, and the worst single rule, the shape of the workload of RQ1, asks how far the objectives diverge.

Graded declaration. Table III shows that heat-aware synthesis never exceeds either baseline on the maximum, which Thm. 23 guarantees, and on these reducts not on the total either, here and at every other p of the sweep. Where synthesis has a choice, the margin shows: its maximum falls below the structure-optimal one on 5 to 7 of the 12 reducts and its total on 7 of them, at every p of the sweep. Against classical synthesis the gain can be the whole retained heat, as on hospital, where structural optimization alone removes it and heat-aware synthesis inherits the removal by Thm. 23. The remaining ties are forced by the reduct: where all three synthesizers tie on the maximum, as on pdbx, the rule that generates the hottest subschema survives in every cover or every alternative is as hot (Lem. 21). The objective also costs little structure. Heat-aware synthesis attains E-BCNF wherever the structure-optimal design does, and its subschema count exceeds the structure-optimal one by at most 4.4% in any draw, coinciding with it on the reducts marked in Table III. The objective costs no time either. Synthesis takes at most 7.2 s on any reduct, and heat-aware synthesis never more than $1.11\times$ the time of either baseline.

TABLE IV: Redundant value occurrences under data-derived heat (RQ2): in the relation (#red, %red of its complete values), in each design, and those of HA that hot rules cause. Omitted are claims and biocase, where every design stores none, and dblp10k, whose mined FDs do not hold on our copy of the relation.

reduct	#red	%red	3NF	SO	HA	HA hot
breast	800	10.4	12	12	12	0
bridges	354	26.7	74	58	54	0
echo	380	24.0	194	144	134	0
hepatitis	1,479	50.4	7,495	7,122	7,028	1,532
hospital	1,608,730	94.1	1,554	0	0	0
ncvoter	2,759	17.1	1,362	917	917	0
pdbx	115,164,324	51.7	67,740,619	67,740,619	67,740,619	67,277,597
routes	201,775	36.3	67,597	67,597	67,597	67,597
weather	2,008,186	46.5	28,151,801	17,313,024	17,313,024	12,664,374



(a) graded declaration ($p = 0.5$) (b) worst single rule

Fig. 3: Maximal heat of the baselines as a multiple of HA's on the 12 reducts (RQ2): (a) SO under the graded declaration; (b) 3NF and SO under the worst single rule, the five reducts without a separating rule omitted. A missing bar is a tie.

Data-derived declaration. Table IV shows that under the data-derived declaration, which no application could state (Sec. III), heat-aware synthesis never stores more redundant occurrences than either baseline, and strictly fewer than the structure-optimal design on three reducts. The same holds for the hot occurrences, those caused by a rule in the top fifth of the scale: the heat-aware design stores none on five of the nine reducts, and no more than either baseline on the other four.

Worst single rule. Fig. 3 shows that the objectives diverge most where the heat concentrates on one rule. Spread over many rules by the graded declaration, heat keeps the optimizing designs within small factors of each other, with SO hotter than HA on half of the reducts (Fig. 3(a)). The worst single rule separates seven of them, by up to $10\times$, the most its scale of ten levels admits (Fig. 3(b)). There the structure-optimal design stores the hot rule as a non-key FD, while heat-aware synthesis stores it under a key of another subschema. On ncvoter it even drives SO above 3NF, the converse of hospital. Hence neither frequency-blind synthesizer dominates the other, and the number of non-key FDs is the wrong measure where the workload concentrates on one rule. On the five reducts marked in Table III no rule separates the two designs, so it reaches every reduct on which synthesis has a choice.

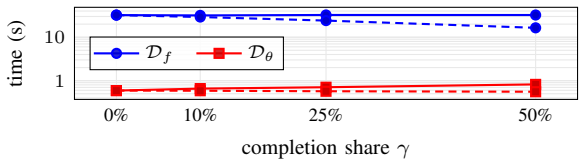


Fig. 4: Mixed workloads at a group of 3,000 (RQ3): time for 200 updates as the completion share γ grows, with conflicting (solid) and clean (dashed) completions.

TABLE V: The delivery designs at 10^6 tuples (RQ3): what they cost to read from and to store.

		$\mathcal{D}_f$	$\mathcal{D}_\theta$	$\mathcal{D}_f/\mathcal{D}_\theta$
scope reconstruction	ms	216	220	0.98×
delivery by key	ms	0.13	0.13	1.00×
history of one courier	ms	43.7	38.3	1.14×
stored size	MiB	95.5	139.2	0.69×

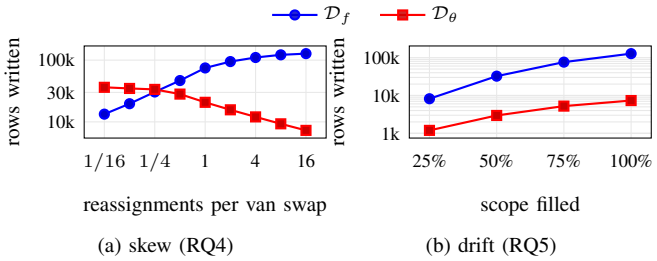
E. Mixed Refresh–Completion Workloads (RQ3)

RQ1 times each operation in isolation. RQ3 interleaves refreshes with completions, and then turns to reads. Fig. 4 shows that completions dilute the advantage on refreshes but do not remove it. The workload of pure refreshes separates the designs by $55\times$. Clean completions cost the same on both designs, so as the completion share γ grows the time on $\mathcal{D}_f$ falls while that on $\mathcal{D}_\theta$ stays flat, and the ratio nearly halves. Conflicting completions instead repair a latent conflict, which costs about what one reassignment costs, so the ratio declines only to $40\times$ and the absolute gap stays where it is.

Table V shows that tailoring does not slow reads. Reconstructing the scope and finding one delivery by its key take about the same time on both designs at 10^6 tuples, since each joins six subschemata of which exactly one is large. The history of one courier runs $1.14\times$ faster on $\mathcal{D}_\theta$, which reads the zones of a courier off the twelve-row subschema cz before it touches the large one. What tailoring costs is storage: $\mathcal{D}_\theta$ occupies $1.46\times$ what $\mathcal{D}_f$ does, because its critical subschema $bdvz$ has three minimal keys against the two of $bcdz$ and pays for one more unique index. Heat counts the non-key FDs of a subschema and not the indexes it carries, so the objective does not see this cost.

F. Heat Skew and the Crossover (RQ4)

The operational studies so far concentrate the workload on one hot operation. RQ4 asks how much of that concentration the advantage needs. Fig. 5(a) shows that skew decides which design is cheaper. The designs cross at about one reassignment per four van swaps: below it $\mathcal{D}_f$ writes fewer rows, and above it $\mathcal{D}_\theta$ does. The two directions are not symmetric. Indeed, a reassignment rewrites the deliveries of a courier, $2k$ rows, while a van swap rewrites those of one branch and courier, $\frac{2}{3}k$ rows, so the advantage grows three times faster in the direction of the courier. At equal rates the window still parts by $3.6\times$, which is that asymmetry and not the skew. Heat crosses later than the rows: it prefers $\mathcal{D}_\theta$ only from equal rates on, since it counts rules and not the rows a rule rewrites.



(a) skew (RQ4) (b) drift (RQ5)

Fig. 5: Robustness (RQ4, RQ5), rows that one window of 1,000 updates writes at a full group of 192: (a) as its traffic moves from van swaps to reassignments; (b) as the scope fills.

TABLE VI: What a declaration costs (RQ5): rows that one window of 1,000 updates writes on the design returned by heat-aware synthesis, as a multiple of the cheaper design's, at a group of 192. A profile is served right where the factor is 1.00.

heat declared	delivery desk	split	fleet office
by the measured rates	1.00×	1.00×	1.00×
uniformly	17.46×	3.63×	1.00×
by the group each rule ranges over	1.00×	1.00×	3.38×
with the two rates exchanged	17.46×	3.63×	3.38×

Between one reassignment per four swaps and equal rates, an exact declaration therefore returns $\mathcal{D}_f$ and pays up to $3.6\times$.

G. Robustness under Drift and Misestimation (RQ5)

RQ1 to RQ4 hold the scope fixed and the declaration exact. RQ5 lets the scope fill and lets the declaration name the wrong operation. Filling the scope from a quarter to full grows what the window writes on both designs, and widens the separation from $6.9\times$ to $17.4\times$ (Fig. 5(b)). Indeed, completeness drift is scope entry (Def. 26) in bulk: every van whose network is registered grows a zone, and with it the redundancy of every rule over it. A design chosen today thus gains value as the data completes, provided the declaration names the right operation.

Table VI shows that misestimated heat costs the whole gain, by the same mechanism that creates it. Of the four declarations, only the one counted from the measured rates returns the cheaper design in all three windows. At the split profile the two refreshes tie and the conflicting completions decide: charged to the courier they return $\mathcal{D}_\theta$, and charged to the van, as the exchange of the two rates does, they return $\mathcal{D}_f$ at $3.63\times$. Declaring uniformly, which counts non-key FDs (Prop. 15), returns $\mathcal{D}_f$ everywhere and pays up to $17.5\times$ at the delivery desk, and declaring by group size returns $\mathcal{D}_\theta$ everywhere and pays $3.4\times$ at the fleet office. Indeed, a reassignment rewrites larger groups than a van swap, yet at the fleet office van swaps arrive fifteen times as often.

H. A Window at Declared Rates (RQ6)

RQ4 and RQ5 count the rows a window writes. RQ6 times the three workloads of Table II live, every kind of update the schema admits in mixed order and at its own rate, as a maintainer meets them.

TABLE VII: One maintenance window of 1,000 updates over 36,864 tuples in groups of 3,072 (RQ6): what it writes and what it takes on each design, under three rate profiles.

	rows written			update time (s)		
window	$\mathcal{D}_f$	$\mathcal{D}_\theta$	ratio	$\mathcal{D}_f$	$\mathcal{D}_\theta$	ratio
delivery desk	1,980,617	41,907	47.26×	54.14	3.02	17.90×
split	1,121,121	258,834	4.33×	32.26	6.39	5.05×
fleet office	138,639	537,797	0.26×	6.46	10.92	0.59×

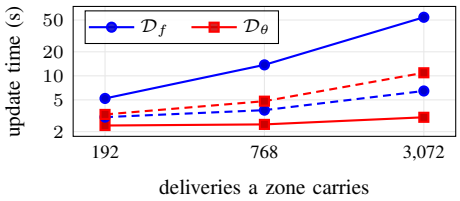


Fig. 6: The same window as the group of a zone grows (RQ6): the desk profile (solid), which reassigns couriers, and the fleet profile (dashed), which swaps vans.

Table VII shows that the profile decides which design is cheaper. At the delivery desk $\mathcal{D}_\theta$ finishes in 3.02 s where $\mathcal{D}_f$ takes 54.14 s. At the fleet office the same two designs change places, because the traffic now refreshes the rule that $\mathcal{D}_\theta$ stores as a non-key FD (Sec. III-B).

Of the three synthesizers, only the heat-aware one follows the profile. The structure-optimal criterion cannot reach $\mathcal{D}_\theta$ at all. Over 200 orders of the atomic closure it returns $\mathcal{D}_f$ or its variant of Ex. 24, the $\mathcal{D}_f$ family, every time, since $bdvz$ carries two non-key FDs against the one of $bcdz$. Classical synthesis, which ranks nothing, returns that family in 102 of the same orders and the $\mathcal{D}_\theta$ family in the other 98, so on this schema the input order alone decides its output. Heat-aware synthesis returns the family the rates point to in all 200 orders, at the delivery desk and at the fleet office alike. Schemata on which HA and SO part in every order are rare: twelve of the fifty thousand we generated over five to seven attributes, among them that of Ex. 1. The rarity limits how often the objective decides, not what it saves where it does.

Fig. 6 shows that where the objective decides, its savings grow with the groups, at both profiles. As the groups grow sixteenfold, the design each profile favors takes at most twice as long, while the other takes three to ten times as long. This is the group rewrite of Sec. III-B in the time domain.

I. A Window Mined from a Real Release (RQ7)

RQ6 shows that the rates decide which design is cheaper, but the rates are ours. RQ7 mines them from a real release. The window spans the editions of 30 June and 19 August 2024, between which country-level attributes are corrected.

Fig. 7 shows that the refreshes of the priced attributes are skewed across the rules of the reduct. The rules of five country-level attributes receive 99% of them, and 728 of 1,614 rules receive none. Indeed, a refresh moves a group of every rule that determines its attribute (Lem. 6). Hence an attribute is equally

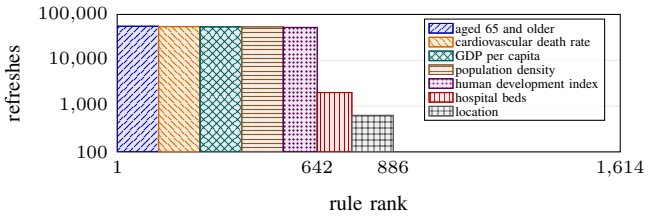


Fig. 7: Refreshes per rule in the window (RQ7): rules ranked by refreshes, colored by the attribute they determine.

TABLE VIII: The window of 19 August 2024 (RQ7): the relation and its priced updates, and on each design the subschemata, the total heat, and what the priced updates cost under the observed and the re-numbered levels. Group rows are written under non-key FDs, and bold marks the least in each line.

the window			
tuples	168,655	tuples in the scope	58,802
atomic FDs of the reduct	1,614	minimal keys	1
refreshes of the 7 priced attributes	270,924	priced updates they coalesce into	409

the designs	*3NF*	*SO*	*HA*
subschemata	133	141	131
total heat	509	6	**4**

the 409 priced updates	*3NF*	*SO*	*HA*
statements	9,546	5,804	**2,056**
rows written	1,667,938	880,116	**369,080**
group rows	170,655	1,830	**0**
refresh time (s)	293.2	113.1	**61.7**

refresh time (s) under re-numbered levels	*3NF*	*SO*	*HA*
ℓ^2, 3,815 priced updates	1,853.0	1,149.9	**587.2**
2^ℓ, 13,575 priced updates	5,703.3	3,072.9	**1,723.0**
ℓ^3, 34,469 priced updates	13,033.8	6,945.0	**4,187.4**

hot under each of its determinants, and a design avoids that heat only by storing none of its rules as a non-key FD.

Table VIII shows that the coolest design, HA, runs the priced updates fastest in every repetition, by 1.83× against SO and 4.75× against 3NF. Fig. 8 shows that the gap persists as the priced updates multiply: across the observed and the re-numbered levels, SO takes 1.66 to 1.96× the time of HA, and 3NF 3.11 to 4.75× (Table VIII).

Part of the gap lies in the group rows, the layer Def. 9 prices, which track the heat (Table VIII). Indeed, each priced update corrects a country-level attribute on every day of a country.

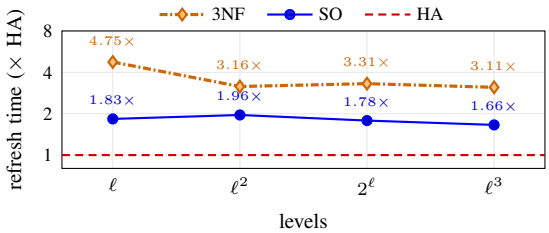


Fig. 8: Refresh time of the priced updates on 3NF and SO as a multiple of HA's (RQ7), under the observed levels ℓ and their re-numberings: the dashed line marks HA.

3NF stores the rule deriving the human development index from the share of people aged 65 and older as a non-key FD in six subschemata, so each priced update of the index rewrites a group in all six. The only hot rule SO stores as a non-key FD derives the hospital beds from the median age and the share of male smokers, and the window corrects the beds in only three priced updates. The rest of the gap between SO and HA lies in the copies of the corrected attributes stored under keys, which heat does not see. SO issues $2.8\times$ the statements of HA and writes $2.4\times$ the rows.

VII. Related Work

Constraints and design under incompleteness. FDs over the incomplete relations of [20], [21], [35] were axiomatized under no-information [36] and weak/strong semantics [37], [38] and surveyed in [39]. Implication under NOT NULL [40], SQL normal forms [24], [41], keys under missing values [27], [28], and cardinality constraints with FDs [42] followed. The design theory of Levene and Loizou [43] applies its FDs to every tuple regardless of missing values and has no notion of frequency, so it takes neither of our two inputs. Or-set expansion into possible worlds [44] and certainty-graded FDs [45] take neither input either. Embedded FDs and UCs, their discovery, and completeness-tailored normalization [16], [17], [46], [47] take completeness as input and are the foundation of this paper.

Updates and queries over incomplete data. Updates of incomplete databases have been studied model-theoretically [48]–[50], whereas our taxonomy is operational. Certain answers over SQL nulls [15], [51], [52] and query completeness [53] concern queries, and the latter marks complete parts of the data, as scopes do.

Normal forms and their justification. Synthesis and its guarantees originate in [19], [30], [31], and its target normal form was refined in [54]. Normal forms have been justified semantically by redundancy and anomalies [18], [22], [25], [26], [55], [56]. Structural parameterizations of BCNF and 3NF [12], [57] and mixed covers [1], [58] optimize update-relevant structure over complete relations, counting the non-key FDs and the minimal keys but not how often a rule is updated. Closest to our work, cardinality-bounded design [13], [14] bounds, for every non-trivial FD, how often a value of its left-hand side occurs, and decomposes by the FDs prioritized for the effort they cause. It grades a design by how many rows one update may touch, and heat grades it by how often its non-key FDs are updated, the two quantities of Sec. I.

Integrity maintenance and repair. Simplified integrity checking [59] and production rules derived from constraints [60] enforce a declared constraint on every write, but take the schema, which decides how many rows each write touches, as given. Repairing violations by value modification is NP-hard already for FDs [61], [62], and repairs can weigh several constraints against one another [63]. In the repair of Prop. 27, the policy of the application fixes which values change, and the design sets how many stored rows that takes.

Workload-driven design at other layers. Access frequencies entered design four decades ago, in affinity-based vertical partitioning [6], and later in physical design advisors [7]–[9], hot-cold classification [2]–[4], and workload-driven partitioning [64], [65]. Vertical partitioning chooses a decomposition from a workload by attribute affinity, not by constraints, so it neither guarantees dependency preservation nor targets a normal form. The others, like the materialized views and partial indexes that speed up access to hot rows [66], tune beneath a fixed schema and inherit the redundancy it admits. Schema design for NoSQL stores [67], [68] and relational denormalization [69] also start from a workload, but add redundancy for reads and preserve no constraints. No prior work tailors logical design to the completeness requirements and the update frequencies of an application at once.

VIII. Conclusion and Future Work

We have introduced completeness- and update-tailored database design. The application declares the completeness requirement under which each business rule applies and how often the rule is updated. The heat of a schema, the total update frequency of its non-key FDs, counts the refreshes that enforcement leaves to triggers rather than keys, and bounding it yields the family of (h, E)-tailored normal forms between E-BCNF and E-3NF. We have characterized the minimum heat of the E-3NF designs that are lossless and dependency-preserving over the scope, and given a synthesis algorithm that attains it. Completions alone surface the conflicts that refreshes leave hidden outside the scope, and the redundancy that prices a refresh also prices their repair. Experiments show that the update frequencies, and not the structure, decide which design is cheaper, and that choosing by structure can cost over an order of magnitude. On a release window of real incomplete data, the tailored design runs the priced updates fastest.

Future work will combine heat with declared bounds on group sizes and address several scopes and other data models. Cardinality-bounded design [14] bounds the group sizes that multiply each heat, and it would be interesting to see which designs attain a heat bound and a cardinality bound at once. An application whose completeness requirements differ between subschemata [17] embeds each subschema in a scope of its own, on which the heat of a rule aggregates the modes embedded there (Lem. 8). One may then ask how synthesis trades heat between scopes. The normal forms for XML [26], [56] and for property graphs [70] also grade a design by its structure alone, and their applications can declare update frequencies in the same way.

IX. AI-Generated Content Acknowledgement

The authors used the AI assistant Claude to polish the language of all sections and to write part of the code behind the experiments of Sec. VI. The authors reviewed all AI-assisted text and code and take full responsibility for the content of this paper.

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APPENDIX

The proofs follow the order in which the results appear in the text.

Proof of Lem. 6. Let $X \to a$ be a non-trivial FD implied by $\Sigma[E]$. Since $a \notin X$, the rewriting leaves every X-value unchanged, so the X-groups of r' are those of r^E. If r' satisfies $X \to a$ and $t \in T$, then every tuple of the X-group of t carries v in r', and belongs to T since v is fresh. Hence T is closed under the X-groups of every such FD, that is, T is a union of components. Conversely, if T is a union of components, every such X-group lies within T or outside it, so in r' it carries v throughout or is unchanged, and $X \to a$ holds as it holds in r^E. □

Proof of Lem. 8. Since $E_i \subseteq E$, every E-total tuple is E_i-total, so $r^E \subseteq r^{E_i}$. An application of mode i with condition x refreshes all tuples of r^{E_i} with X-value $x[X]$, hence, restricted to the stored scope, exactly the tuples of r^E with X-value $x[X]$. This set does not depend on i, so the applications of all m modes rewrite the same stored rows, and their frequencies add up on those rows. □

Proof of Prop. 11. The projected constraints of a schema S over R imply exactly the FDs over S that $\Sigma[E]$ implies, so the atomic closure of S consists of the members of $\mathcal{A}$ inside S. A member $X \to a$ inside S therefore lies inside S', and if X is no superkey of S, it is no superkey of $S' \supseteq S$. Hence every non-key atomic FD of S is a non-key atomic FD of S', and since heats are sums of positive numbers over these sets, $\mathcal{H}_\theta(S) \leq \mathcal{H}_\theta(S')$. □

Proof of Prop. 13. A schema (R_i, Σ_i) is in E-BCNF if and only if every member of $\mathcal{A}_i$ has a superkey determinant, since every non-trivial implied FD has a member of $\mathcal{A}_i$ with the same right-hand side and a determinant inside its own. This holds if and only if the sum in Def. 9 is empty. An empty sum is 0, and a non-empty one is positive, since θ_E takes positive values. Since E-BCNF implies E-3NF, heat 0 is the same as $(0, E)$-TNF. □

Proof of Prop. 14. The nesting is immediate, and the heat of any schema is at most $\mathcal{H}_{\text{tot}}$, so the top level imposes no bound beyond E-3NF. For strictness between levels $h < h'$, take $R = \{a, b, c\}$ with $\Sigma = \{(R;\, ab \to c),\ (R;\, c \to b)\}$, scope $E = R$, and $\theta(R;\, c \to b) = h'$. The minimal keys are ab and ac, so every attribute is prime and the schema is in E-3NF. Its atomic closure is $\{ab \to c,\ c \to b\}$, of which only $c \to b$ is non-key, so $\mathcal{H}_\theta(R) = h'$. Hence the schema is in (h', E)-TNF but not in (h, E)-TNF. □

Proof of Prop. 15. Under constant $\theta_E \equiv w$ the heat of every schema is w times its number of non-key atomic FDs, and maximizing or minimizing over subschemata scales by the constant w. □

Proof of Prop. 16. In the delivery schema of Ex. 1, set $\theta(cgz;\, z \to c) = 2L$ and all other heats to 1. The atomic closure and the minimal keys of R are as listed in Ex. 10. Each decomposition of Fig. 1 has six subschemata and leaves eight of the 18 members of $\mathcal{A}$ unembedded, all implied by the members it embeds. Seven of them have two-attribute determinants, such as $bc \to g$, implied by $bc \to v$ and $v \to g$, and $rz \to b$, implied by $rz \to v$ and $vz \to b$. The eighth is a zone rule: $bdv \to z$ lies in no subschema of $\mathcal{D}_f$ and is implied by $bv \to c$ and $bcd \to z$, and $bcd \to z$ lies in none of $\mathcal{D}_\theta$ and is implied by $bc \to v$ and $bdv \to z$. Both decompositions are therefore E-dependency-preserving, and both are E-lossless by the classical synthesis argument on the scope [31], since each contains a minimal key of R, namely $bcd \subseteq bcdz$ and $bdz \subseteq bdvz$. Five subschemata per decomposition are in E-BCNF. The remaining ones are $bcdz$, with the non-key FD $z \to c$, and $bdvz$, with the non-key FDs $bz \to v$ and $vz \to b$ (Ex. 10), and both are in E-3NF, since c, v, and b lie in the minimal keys bcd and bdv of their subschemata. Hence $\mathcal{D}_f$ materializes one FD redundantly and $\mathcal{D}_\theta$ two. None of this depends on θ, so it holds for every L. The heats of $\mathcal{D}_f$ and $\mathcal{D}_\theta$ are $2L$ and 2, so the design that stores the single non-key FD is the hotter one, and the separation is the factor L. □

Proof of Prop. 19. Suppose some $X \to a \in \mathcal{A}_i$ is non-key in R_i, and let X^+ be the closure of X in (R_i, Σ_i). Take the two-tuple instance whose tuples agree exactly on X^+ and differ elsewhere. The tuples are distinct because X is not a superkey. This instance satisfies Σ_i, since whenever both tuples agree on a determinant Y of an implied FD $Y \to b$, we have $Y \subseteq X^+$, hence $b \in X^+$, so they agree on b. In particular, no minimal key lies inside X^+, as it would make X a superkey. Applying the mode $X \to a$ with a fresh value to one tuple alone now violates $X \to a$ itself, while rewriting both tuples restores agreement on X^+ and hence satisfaction, so the application is anomalous.

Conversely, let the schema be in E-BCNF, so every determinant of $\mathcal{A}_i$ is a superkey of R_i, and consider an application of a mode $X \to A$. Then X itself is a superkey, and two distinct tuples of a consistent instance never agree on a superkey, so the X-group is a single tuple and single-tuple and whole-group rewrites coincide. The rewrite violates nothing either, since a fresh value creates no new agreement between distinct tuples, so no two of them come to agree on a determinant of $\mathcal{A}_i$. □

Proof of Thm. 20. For hardness, let θ_E be constant with value c and let $h = c \cdot |R_i| \cdot 2^{|R_i|}$, a number of polynomially many bits. Since $|\mathcal{A}_i| \leq |R_i| \cdot 2^{|R_i|}$, the bound h is at least the total heat of $\mathcal{A}_i$, so membership coincides with E-3NF (Prop. 14). E-3NF is 3NF of the projected constraints, which is NP-hard to recognize by the classical primality argument [17], [29].

Now let $\mathcal{A}_i$ be computable in polynomial time. Then $\mathcal{H}_\theta(R_i)$ is computed by one attribute closure per member of $\mathcal{A}_i$, which decides whether its determinant is a superkey of R_i, followed by a sum.

For membership in NP, the verifier computes $\mathcal{A}_i$ and $\mathcal{H}_\theta(R_i)$ as above and compares the heat against h. It remains to verify E-3NF, which holds if and only if every non-key member of $\mathcal{A}_i$ has a prime right-hand side, since every non-trivial implied FD has a member with the same right-hand side

and a determinant inside its own. The certificate supplies, for every member $X \to a$ whose determinant an attribute closure shows to be no superkey, a minimal key of R_i containing a. Each supplied key is verified to be a key by one closure and to be minimal by dropping each attribute in turn. The verifier accepts if and only if every such right-hand side lies in a verified minimal key, and such keys exist exactly when the schema is in E-3NF, so verification is sound and complete.

For the interreducibility, E-3NF reduces to determination on the same input by taking θ_E constant and h the bound above. Conversely, determination reduces to one E-3NF test followed by the polynomial heat computation and the comparison against h.

For the design problem, let θ_E be constant with value 1 and $h = 0$. A schema has heat 0 if and only if none of its atomic FDs is non-key, that is, if and only if it is in E-BCNF (Prop. 13), so the question is whether an E-lossless, E-dependency-preserving decomposition into E-BCNF exists. For $E = R$ this is the existence of a lossless, dependency-preserving decomposition into BCNF, which is NP-hard [19]. □

Proof of Lem. 21. Let $\mathcal{D}$ be such a decomposition, and let $F^{\mathcal{D}}$ collect the members of $\mathcal{A}$ whose attributes lie inside some subschema of $\mathcal{D}$. Every non-trivial implied FD $Y \to a$ whose attributes lie inside a subschema has a member $X \to a$ of $\mathcal{A}$ with $X \subseteq Y$, which lies inside that subschema too. Hence the projected constraints of the subschemata imply exactly what $F^{\mathcal{D}}$ implies, and E-dependency-preservation yields that $F^{\mathcal{D}}$ implies $\mathcal{A}$. Every $X \to a \in F^{\mathcal{D}}$ lies inside some subschema R_i of $\mathcal{D}$, so $\mathcal{H}_\theta(R_i) \geq \mathcal{H}_\theta(Xa)$ by Prop. 11. Taking maxima, the maximal subschema heat of $\mathcal{D}$ is at least $\max_{X \to a \in F^{\mathcal{D}}} \mathcal{H}_\theta(Xa)$, which is at least h^* since $F^{\mathcal{D}}$ is one of the sets the minimum ranges over. □

Proof of Thm. 23. Both elimination passes drop an atomic FD only when the surviving working set still implies it, so $\mathcal{A}_w$ remains a cover of $\mathcal{A}$ throughout. Every member of the final $\mathcal{A}_w$ is embedded in the subschema built for it in line 6 or line 8, so the output is E-dependency-preserving. Line 9 removes only subschemata inside others, which keeps their FDs embedded and cannot raise the maximal heat. Each (Xa, Σ_{Xa}) is in E-3NF, by the classical argument applied to the reduct. For a key subschema added in line 11, a minimal key K admits no implied FD $Y \to b$ with $Yb \subseteq K$ and $b \notin Y$, since $K \setminus \{b\}$ would then determine b and hence all of R, contradicting minimality. That subschema therefore carries no non-trivial constraint, is in E-BCNF, and has heat 0. The output is E-lossless, since it is E-dependency-preserving and, after line 11, some subschema is a superkey of $(R, \Sigma[E])$, either one retained from line 9 or the added minimal key. The classical synthesis argument thus applies to the scope, a total relation satisfying $\Sigma[E]$ [31], [71], and the tuples outside the scope are stored in the full-width remainder [17].

For optimality, let $\mathcal{D}'$ be any decomposition obtained from $\mathcal{A}$ by eliminating redundant atomic FDs, and let h' be its maximal subschema heat. Such a decomposition carries one subschema Xa per retained FD $X \to a$, so h' is the largest heat of these schemata. Suppose the output kept a subschema (Xa, Σ_{Xa}) with $\mathcal{H}_\theta(Xa) > h'$. A positive heat means that (Xa, Σ_{Xa}) is not in E-BCNF, so $X \to a$ belongs to $\mathcal{C}$ and line 6 visits it. Since $\mathcal{D}'$ has no subschema of heat above h', it eliminated $X \to a$, so the members of $\mathcal{A}$ that $\mathcal{D}'$ retains imply it, and each of them has heat at most h'. Line 6 visits $X \to a$ before every FD whose schema is cooler, so at that moment $\mathcal{A}_w$ still contains every FD whose schema has heat at most h' and hence implies $X \to a$. The test succeeds and removes $X \to a$, contradicting its presence in the output. The maximal subschema heat of the output is thus at most h' for every such $\mathcal{D}'$. Every $F \subseteq \mathcal{A}$ that implies $\mathcal{A}$ is the retained set of such a decomposition, since eliminating the members of $\mathcal{A} \setminus F$ in any order succeeds while F is retained, and that decomposition has maximal subschema heat $\max_{X \to a \in F} \mathcal{H}_\theta(Xa)$. Hence the output has maximal subschema heat at most h^*, and at least h^* by Lem. 21.

The secondary claim repeats the argument on line 8, with the number of minimal keys in place of the heat. Let $\mathcal{D}'$ be obtained by eliminating redundant atomic FDs from the working set that the first pass leaves, let n' be the maximal number of minimal keys over its E-BCNF subschemata, and suppose the output kept an E-BCNF subschema (Xa, Σ_{Xa}) with more than n' of them. Then $\mathcal{D}'$ eliminated $X \to a$, so the FDs it retains imply it. Each of those is either critical, and line 8 never visits it, or E-BCNF with at most n' minimal keys, and line 8 visits it after $X \to a$. All of them are therefore still in $\mathcal{A}_w$ when the test on $X \to a$ runs, so the test succeeds and removes $X \to a$, again contradicting its presence in the output. □

Proof of Prop. 25. The atomic closure of a schema Xa consists of the members of $\mathcal{A}$ inside Xa (proof of Prop. 11), so the E-BCNF test of a member of $\mathcal{A}$ and the heat evaluation on line 4 each take one attribute closure per member inside Xa. The two elimination passes perform one implication test per member of $\mathcal{A}$, each an attribute closure. For the ordering of the second pass, let (Xa, Σ_{Xa}) be in E-BCNF. The determinant of every member of $\mathcal{A}$ inside Xa is a superkey of Xa, and a minimal one, since a proper subset that is a superkey would determine the right-hand side and contradict the minimality of the determinant. Conversely, a minimal key K of Xa is a proper subset of Xa, since X already is a superkey, so K determines some attribute outside K, and that FD is atomic, since a proper subset of K that determines the attribute would be a superkey by E-BCNF and contradict the minimality of K. The minimal keys of Xa are therefore the determinants of the members of $\mathcal{A}$ inside Xa, at most $|\mathcal{A}|$ many, and the ordering is a count over $\mathcal{A}$. Line 9 is a pairwise containment scan, the test before line 11 takes one attribute closure per subschema, and line 11 computes one minimal key by removing attributes from R while the remaining ones still determine R. All steps are polynomial in $|\mathcal{A}|$ and $|R|$. □

Proof of Prop. 27. Under newcomer-wins, after the entry all stored tuples of the group must carry $t(A)$. A subschema on

which $X \to A$ is non-key stores, for the group, one row per distinct projection of the group's tuples, each carrying the old A-value, and all of them must be rewritten. If the subschema separates the group's tuples, that is, if these projections are pairwise distinct, the count is k. If every subschema containing XA has X as a key, each of them stores the group in a single row, and rewriting that one row in each restores $X \to A$. The other rules that determine A carry their groups along (Lem. 6), again one row per group wherever such a rule is stored under a key. □

Proof of Prop. 29. (1) A retraction moves no tuple into the scope, so no pending tuple gains a scope tuple to disagree with. If the retracted attribute belongs to XA, the retracted tuple is no longer XA-total and carries no latent conflict with $X \to A$. Otherwise its XA-values are unchanged, and a retraction that moves it out of the scope leaves it agreeing on A with its X-group, since r satisfies the constraint there. (2) The mode rewrites every tuple of its scope $r^{E'}$ in the group, pending or not, and these keep agreeing with the moved group. An XA-total pending tuple that agreed with the group but is not E'-total keeps the old A-value, so once the group moves to a fresh value, it disagrees with the scope and carries a latent conflict. (3) Refreshes, retractions, and inserts outside the scope never move a tuple into the scope, and a tuple inserted into the scope has no pending past, so a latent conflict can surface only through the completion that makes its carrier E-total, whose writes Prop. 27 counts. (4) A tuple inserted into the scope is E-total, so every constraint of the scope applies to it at once, and Prop. 27 counts the writes of the newcomer-wins repair. A tuple inserted outside the scope is invisible to the constraints of the scope, so it violates none, and its values are as inserted, so Def. 28 applies exactly when it is XA-total and disagrees on A with its X-group. □